From news feeds to signals

Ongoing research on complementing business-cycle surveys with news-based indicators

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Based on joint work with Nabil Bouamara, Arthur Jacobs, Viktor Proesmans, Yannick Rombauts, and Ine Steyls

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NationalBank OF BELGIUM
Eurosystem

My talk is about the land of fries and chocolate

Why is Belgium an interesting case for this question?

- small and open
- internally diverse
- and often subject to a strong public narrative about “how Belgium is doing”

Taken together, this makes Belgium **difficult to track in real time**, and a useful test case for complementary text-based indicators.

The task: tracking the Belgian economy in real time

The NBB **needs timely, reliable signals** on the state of the Belgian economy

- Economists continuously parse data from **official statistics, surveys, and other sources**
- The challenge is to **separate signal from noise** in real-time
- This is essential for **business-cycle assessment, short-term forecasting, and communication**

Nowcasting (i.e., forecasting the present of very near future) at the NBB

Our **nowcasting** models combine information from multiple sources to form an early assessment of **current-quarter GDP growth**.

- They extract signal from noise incoming information
- Their inputs include official statistics, surveys, and other short-term indicators
- Focus today: the role of survey-based information, and whether text can complement it

R2D2 is a nowcasting model used at the NBB

Source: Business Cycle Monitor | nbb.be; Het gebruik van BREL om de Belgische conjunctuur te nowcasten: de rol van enquêtegegevens | nbb.be; Presentation of the National Bank of Belgium's Business Cycle Monitor

At the NBB, we collect these inputs (intelligence on economic actors, if you will) through different channels

Business survey (since 1954)

- ~3,000 firms surveyed each month
- Topics: production, order books, employment, prices

Consumer survey (since 1970s)

- ~1,850 households surveyed each month
- Topics: financial situation, unemployment, saving, major purchases

So you're a bit like James Bond, but for the economy?

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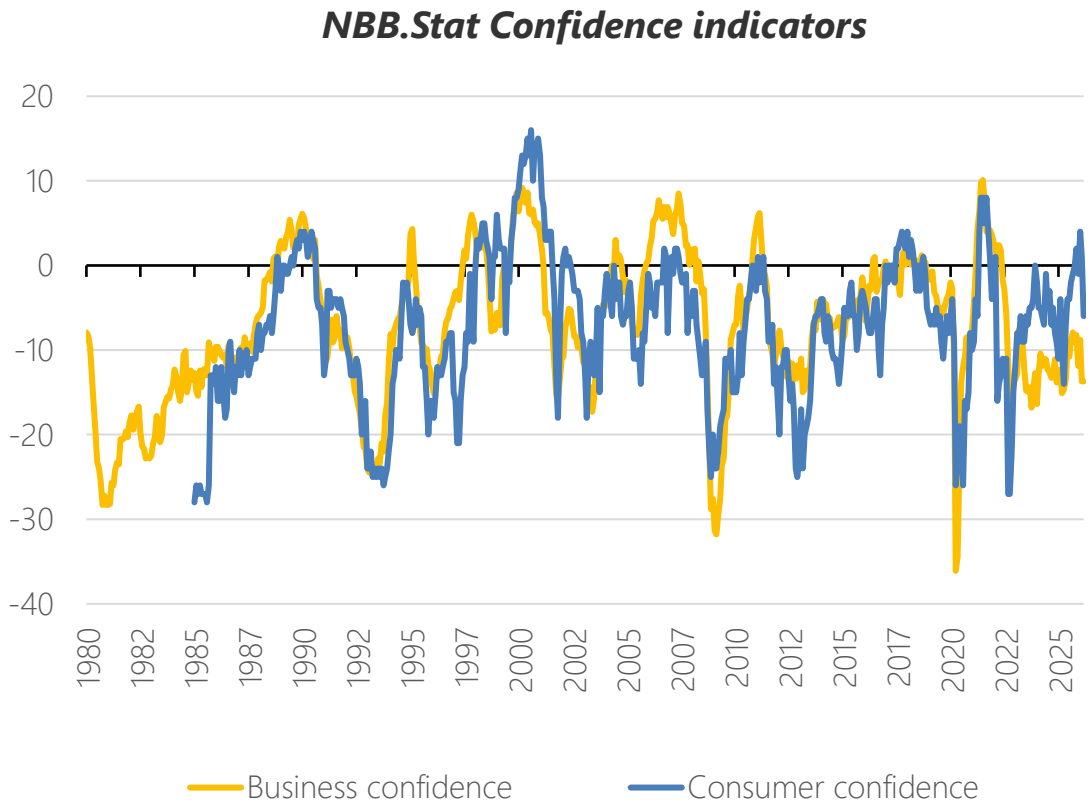
Respondents typically choose something like better / same / worse, and indicators are published as a net-balance

- **= % favourable answers minus % unfavourable answers**



Methodology | National Bank of Belgium
NBB .STAT Data Explorer • Monthly business surveys
NBB .STAT Data Explorer • Consumer survey

Why surveys remain so valuable

Surveys are not just data sources, they are **trusted statistical instruments**

- Direct measurement of perceptions and expectations
- Early signals of turning points
- Additional information beyond standard macro-statistics
- European harmonized design and cross-country comparability
- Rich disaggregation
- Long historical series
- Institutional trust and wide use

Part of the Joint Harmonised EU Programme of Business and Consumer Surveys User Guide

Why look beyond the as-is?

When standard surveys are not enough

- **Limited flexibility:** harmonized questionnaires cannot easily be adapted to new events or fast-changing circumstances
- **Little narrative detail**: net balances are useful signals, but they do not tell us the story behind the numbers
- **Response rates under pressure:** lower participation can weaken representativeness and signal quality over time

Do other timely information sources, including text-based ones, can help fill some of these gaps in a more systematic way?

Ad hoc monitoring in crisis episodes: ERMG and follow-up surveys

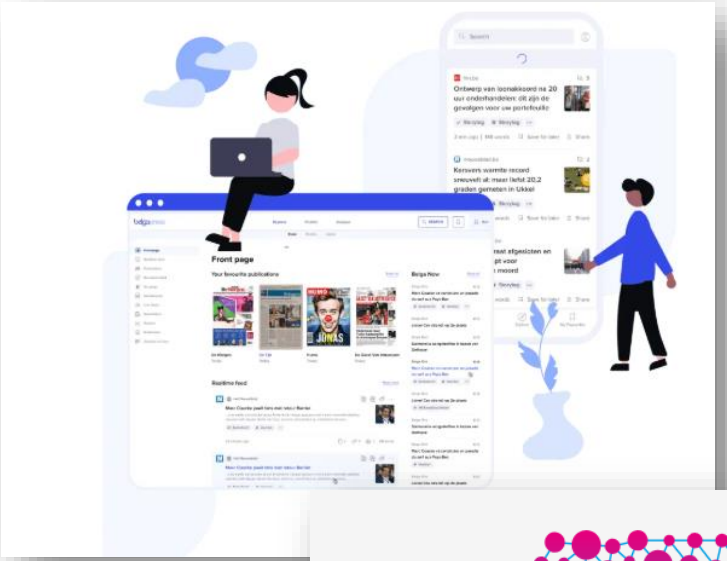
A natural complement: news-based text indicators

News as a first prototype within a broader web-data agenda

Can the news help fill some of these gaps in a more systematic way?

- It is **timely, high-frequency, and available at scale**
- It captures the **economic narrative i**n near real time
- It may help to:
  - flag turning points earlier
  - complement survey signals
  - explain why confidence indicators move
- In our case, we work with news-based indicators constructed by Sentometrics, linked to BelgaPress
  - daily series, available since 1999

Caveat: news is our current prototype, but web data is the broader methodological horizon



sentometrics

How a classical news-based text indicator is constructed

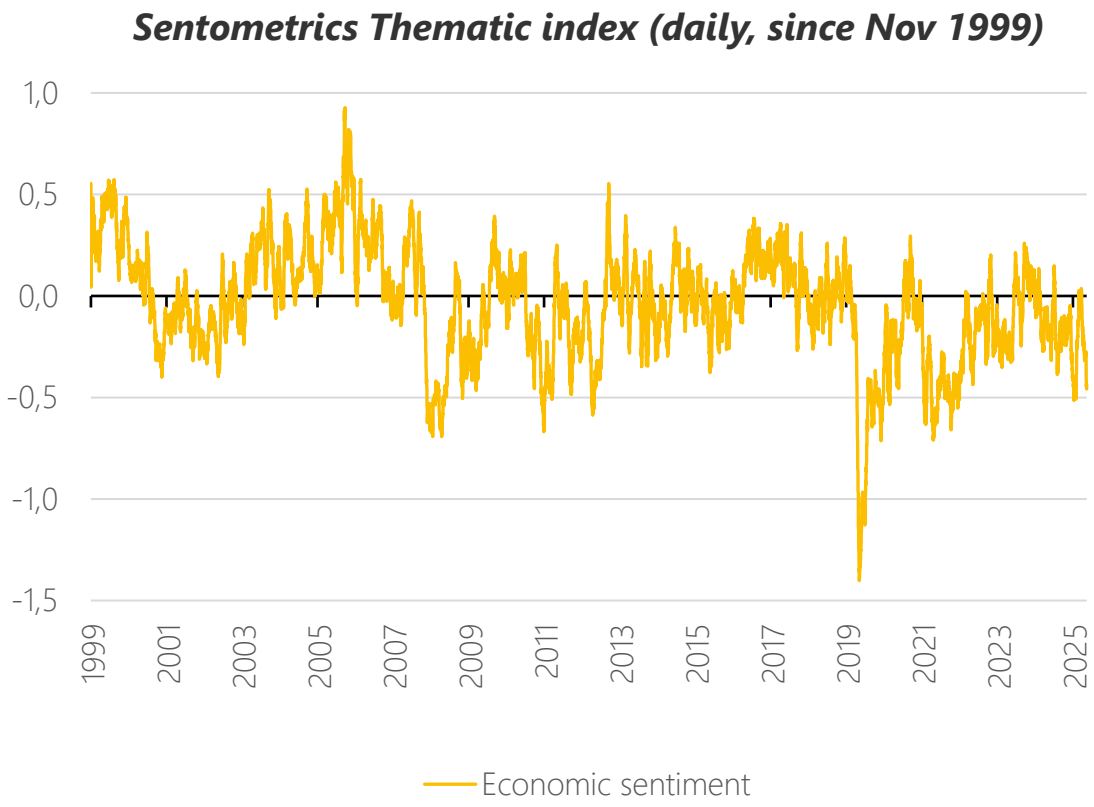
It is built from text, not survey responses.

Sentometrics constructs a thematic indicator, and the general logic is:

1. *start from a stream of news articles*
2. *keep the articles that are relevant to the economy using keywords ('Economy', 'Recession', ...)*
3. *assign sentiment scores to the article, with lexicon-based methods*
4. *aggregate article-level scores over time in a daily series*

Difference with survey indicators

- *Survey indicator: direct answers to fixed questions*
- *News-based indicator: indirect proxy derived from published text*

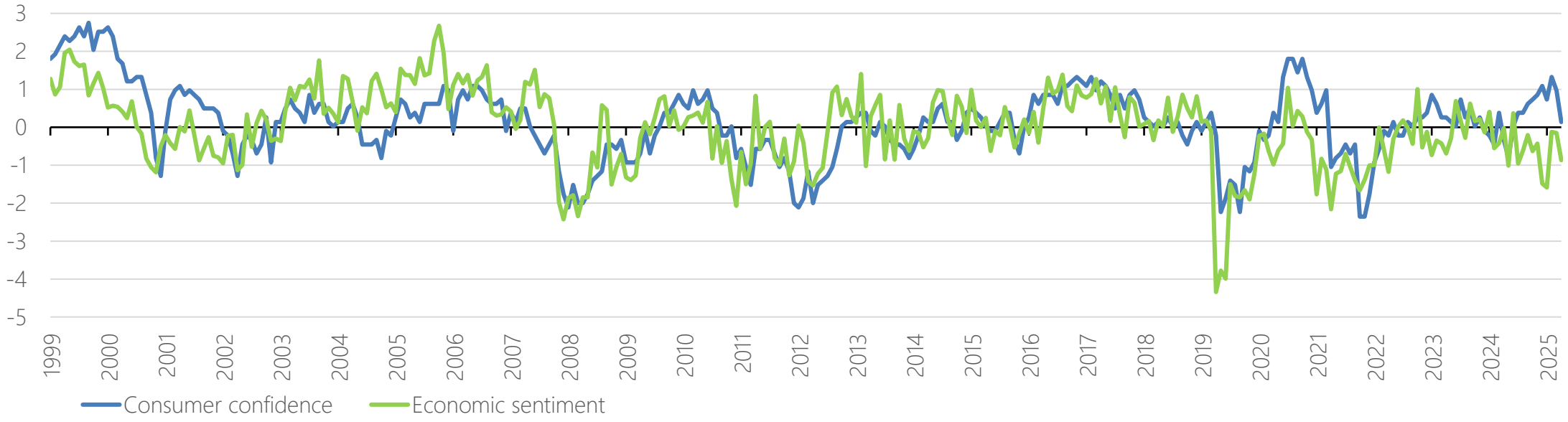


WP14.pdf
Thematic Indicators - Live browser | Sentometrics

NBB - Public

A first prototype

Consumer survey vs. news-based index (since 1999, monthly frequency, standardized)



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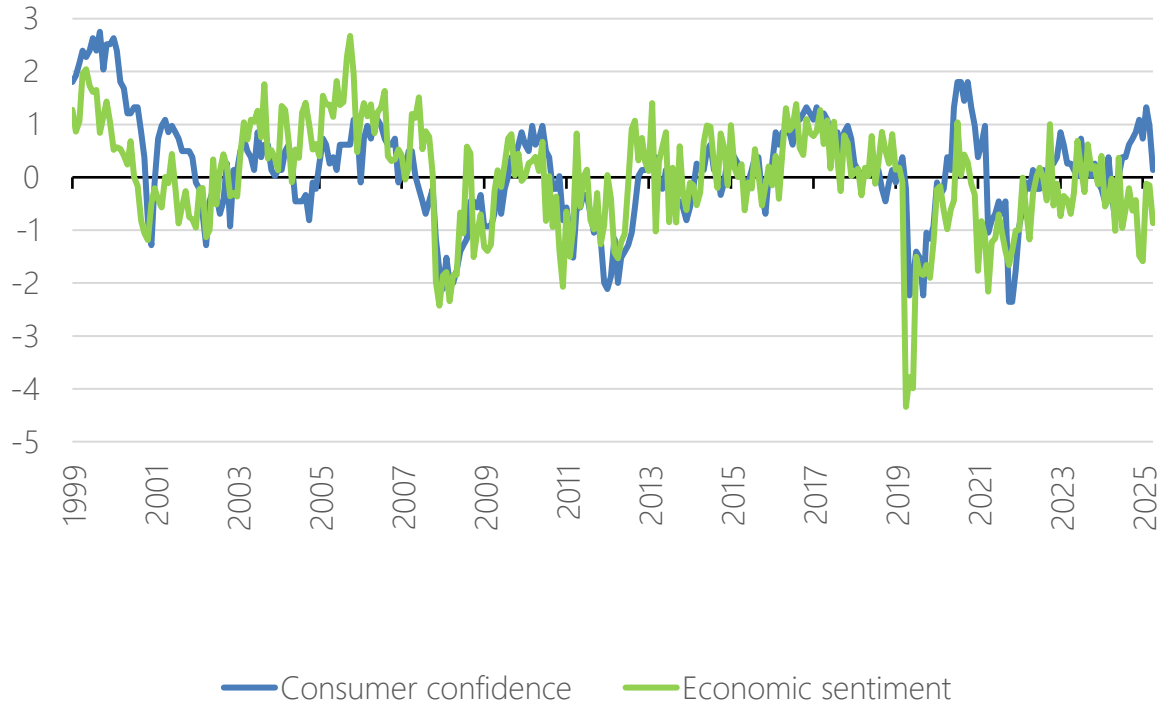

Reasonable co-movement (Pearson *r* = 0.59). Improvements possible via

- better weighting (industry and regional representation),
- LLM-based semantics (beyond word counts)

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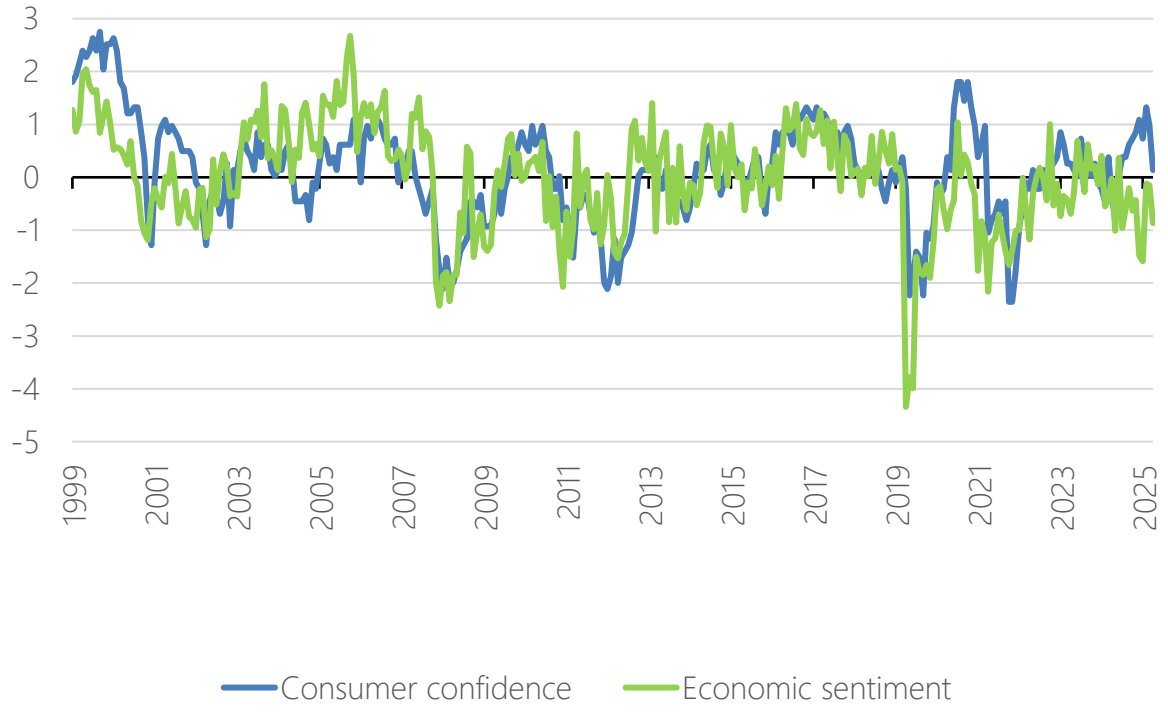

But **news-based sentiment has downsides**;

- we're no longer questioning the households directly, but we're measuring the media **narrative**
- we no longer have the same level of **disaggregated** breakdowns into characteristics
- **cross-country comparability** is harder to ensure than in harmonized survey data

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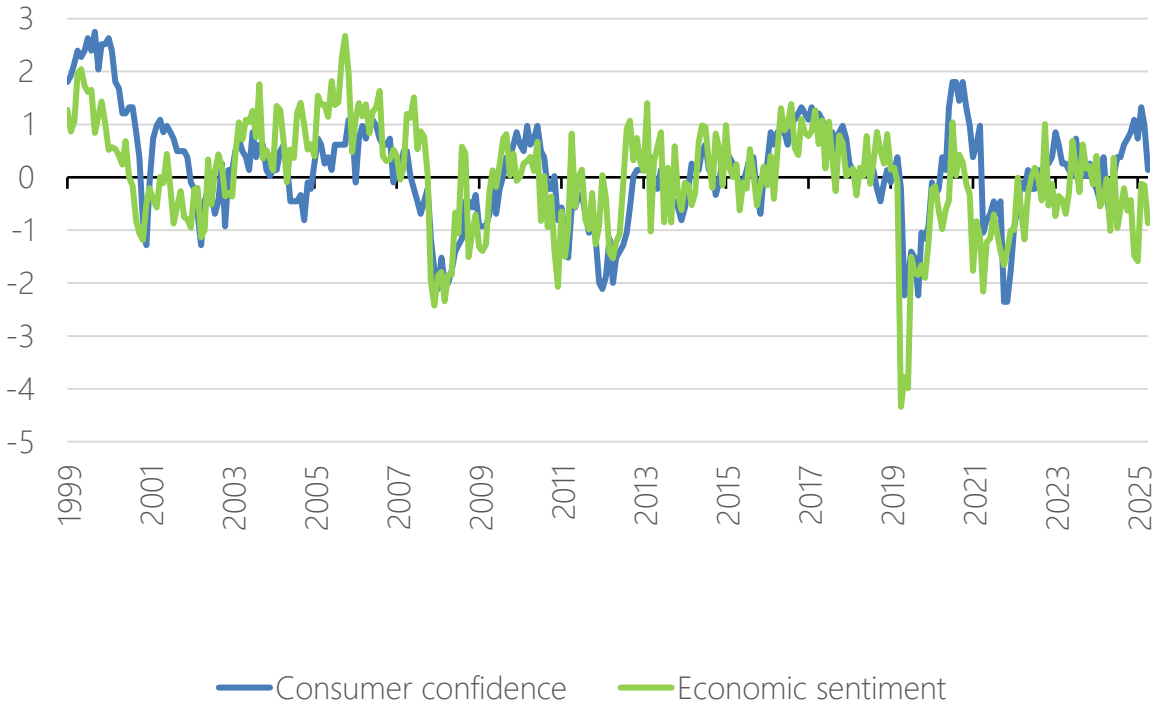

Then again, **consumer surveys also have downsides,**

- Call centers are **expensive**
- declining response rates can **erode data quality** and representativeness
- recent responses may reflect perceptions of government measures, **not only by households' own situation**
- they may **understate the severity** of 'extreme' episodes

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On the use of new/external data sources

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UNITED NATIONS ECONOMIC COMMISSION FOR EUROPE

CONFERENCE OF EUROPEAN STATISTICIANS

UNECE Machine Learning for Official Statistics Workshop 2023

04 April 2023

5-7 June 2023, Geneva Switzerland

Changing Data Sources in the Age of Data Science for Official Statistics

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Abstract

Data science has become increasingly essential for the production of official statistics, as it enables the automated collection, processing, and analysis of large amounts of data. With such data science practices in place, it enables more timely, more insightful and more flexible reporting. However, the quality and integrity of data-science-driven statistics rely on the accuracy and reliability of the data sources and the techniques that support them. In particular, changes in data sources are inevitable to occur and pose significant risks that are crucial to address in the context of data science for official statistics.

This paper gives an overview of the main risks, liabilities and uncertainties associated with changing data sources in the context of data science for official statistics. We provide a checklist of the most prevalent origins and causes of changing data sources; not only on a technical level, but also regarding ownership, ethics, regulation, public perception, etc. Next, we highlight the repercussions of changing data sources on statistical reporting. These include technical effects such as concept drift, bias, availability, validity, accuracy and completeness, but also the neutrality and potential discontinuation of the statistical offering. We offer a few important precautionary measures, such as building robustness in both data sourcing and statistical techniques, and thorough monitoring. By doing so, official statistics based on data science can maintain their integrity, reliability, consistency and relevance in policymaking, decision-making, and public discourse.

Conclusion and next steps

Can web-based sources complement official survey infrastructure for business-cycle monitoring?

- **We believe they do.** But this is only a **first step** in a broader exploration of non-traditional web-based data for official statistics and cycle monitoring.

Beyond news, **other web-based signals** worth exploring include:

- e.g., job postings, firm websites/vacancy pages, retail prices collected online, Google Trends / search intensity, consumer reviews, mobility/traffic/booking indicators, social media

What do you think?